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|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4(a) | <p>1. (deliberate) manipulation / changing, of, DNA / genetic material / genome ;</p> <p>2. to, modify / improve / add, (named) characteristics of, organism / crop / plant ;</p> <p>3. <i>ref. to</i> add / delete / replace, gene / allele (into crop)<br/> <b>or</b><br/> gene editing ; <b>A</b> CRISPR</p> | <b>3</b> |
| 4(b) | <p>any answer between 295 and 305 ;</p> $\frac{(170 - 42.5)}{42.5} \times 100$ <p><i>If incorrect answer max 1 mark for this equation</i></p>                                                                                                                                                                   | <b>2</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 4(c)     | <p>any <b>five</b> from:</p> <ol style="list-style-type: none"> <li>1 <i>idea of</i> increase / high, productivity / yield / food quantity ; <b>A</b> named example</li> <li>2 improve (named) quality ;</li> <li>3 <i>ref. to</i> herbicide / weedkiller, tolerance / resistance ;</li> <li>4 reduces / no, competition from weeds<br/><b>or</b><br/>so crops get more, light / water / AW ;</li> <li>5 <i>ref. to</i> insect / pest, resistance ;</li> <li>6 less / no, crop damage<br/><b>or</b><br/>less food wastage ;</li> <li>7 <i>ref. to</i> economic advantage ;</li> <li>8 tolerance to name factor ; e.g. disease / environmental extreme / frost / high temperatures / windy / less fertile soil</li> </ol> | 5     |